

EDUCATION

Cornell University Ph.D. in Computer Science , Programming Languages Advisors: Prof. Alexandra Silva, Prof. Andrew C. Myers	Ithaca, NY 2025 – Present
University of Pennsylvania, Penn Engineering & The Wharton School M.S.E. in Computer and Information Science B.S.E. in Computer Science , Mathematics minor B.S. in Economics GPA: 3.97/4.00, <i>Summa Cum Laude</i> Research advisors: Prof. Steve Zdancewic, Prof. Sanjeev Khanna	Philadelphia, PA May 2025

SELECTED PUBLICATIONS

Hanxi Chen, Noam Zilberstein, Andrew C. Myers, Alexandra Silva. Oblivious Probabilistic Outcome Logic: Verifying Probabilistic Programs with an Oblivious Adversary. arXiv preprint, 2026.

Calvin Beck, **Hanxi Chen**, Steve Zdancewic. Vellvm: Formalizing the Informal. CoqPL'25, 2025.

Calvin Beck, **Hanxi Chen**, Steve Zdancewic. Vellvm: Formalizing the Informal LLVM (An Experience Report). NFM'25, 2025.

Calvin Beck, Irene Yoon, **Hanxi Chen**, Yannick Zakowski, Steve Zdancewic. A Two-Phase Infinite/Finite Low-Level Memory Model: Reconciling Integer–Pointer Casts, Finite Space, and undef at the LLVM IR Level of Abstraction. ICFP'24, 2024.

RELEVANT RESEARCH EXPERIENCE

Research Assistant , <i>Cornell PL Group</i> , advised by Prof. Alexandra Silva and Prof. Andrew C. Myers	Sep 2025 – Present
Cornell University , Computer Science - Developed Oblivious Probabilistic Outcome Logic (opOL), a logic for probabilistic nondeterministic programs with oblivious adversaries; used schedule consumption and probabilistic independence to verify caching and leader-election protocols. - Mechanized opOL in Lean 4, formalizing monadic semantics, probability-space constructions, probabilistic assertions, semantic and syntactic outcome triples, soundness proofs, and verified case studies. - Contributed Lean mechanization for Type-Confusion Security, formalizing information-flow checks for decentralized systems where untrusted components can lie about types and induce confused-deputy attacks.	
Research Mentor , <i>Cornell Math+AI</i> , advised by Prof. Daniel Halpern-Leistner	Jan 2026 – Present
Cornell University , Mathematics Mentoring and building an autoformalization benchmark for translating research-level mathematical statements and proofs into Lean 4; curating problem/proof pairs and co-developing proof-quality evaluation and LaTeX-to-Lean tools.	
Research Assistant , <i>PLClub</i> , advised by Prof. Steve Zdancewic	Jun 2021 – Aug 2025
University of Pennsylvania , Computer and Information Science - Contributed to the ICFP 2024 Vellvm memory-model paper in Prof. Steve Zdancewic's group, helping design a low-level memory model for LLVM optimization reasoning, verifying serialization lemmas in Rocq, and implementing tests for the memory model. - Built Rocq/OCaml validation tools for testing Vellvm's semantic faithfulness against LLVM implementations, including QuickChick generators for UB-free LLVM programs, interpreter-based traces, and a differential testing framework that found novel LLVM bugs. - Developed Rocq infrastructure for permutation reasoning and list AC unification, then applied it to formalize first-order classical linear logic with cut elimination and a linear-resource calculus for asynchronous queries.	
Research Assistant , <i>Wharton Research Scholars</i> , advised by Prof. Sanjeev Khanna	Jan 2023 – Dec 2024
University of Pennsylvania , The Wharton School - Studied precedence-constrained knapsack problems, which are strongly NP-hard in general, and proved pseudo-polynomial algorithms for structured poset classes including unions of three paths and one-directional bipartite posets with consecutive children. - Presented the work in the Wharton Research Scholars seminar and wrote an undergraduate thesis.	

SELECTED TECHNICAL PROJECTS

RISC-V Pipelined Processor – <i>SystemVerilog, Python, FPGA</i>	Feb 2025 – May 2025
University of Pennsylvania , <i>CIS 5710 Computer Organization and Design</i> - Implemented a RISC-V pipelined processor with AXI-Lite-style caching, multi-cycle units, branch prediction, and memory-mapped I/O. - Built verification testbenches, optimized LUT usage and timing behavior, and synthesized the design to FPGA to run compiled programs.	
Oat Compiler – <i>OCaml, C, LLVM IR, x86, Clang, Dune</i>	Feb 2024 – May 2024
University of Pennsylvania , <i>CIS 7000 Compilers and Interpreters</i> - Implemented a compiler for Oat, a C-like language with pointers and structured data, including lexing, parsing, type checking, frontend lowering, and backend code generation to LLVM IR and x86. - Added static analyses and optimizations including mem2reg and dead-code elimination, validated against thousands of test cases.	

SKILLS & RESEARCH INTERESTS

Programming & Tools: Lean 4, Rocq/Coq, OCaml, Haskell, C/C++, Python, PyTorch, Java, Rust, SystemVerilog, LLVM IR/Clang, Z3, Docker, property-based testing, static analysis, benchmarking.

Research Interests: compilers, formal verification, program logics, probabilistic programming, concurrency, information flow.